

Van Crate Capacity Calculation

Van dimensions: Length 4.85 m, Width 2.3 m, Height 1.8 m.

Crate dimensions: Length 0.7 m, Width 0.5 m, Height 0.3 m.

Best practical orientation:

- 0.5 m along van length: $\text{floor}(4.85/0.5)=9$
- 0.7 m along van width: $\text{floor}(2.3/0.7)=3$
- 0.3 m along van height: $\text{floor}(1.8/0.3)=6$

Total crates = $9 \times 3 \times 6 = \mathbf{162 \text{ crates}}$.

Volume check:

Van volume = 20.079 m^3 .

Crate volume = 0.105 m^3 .

Theoretical volume limit = about 191 crates, but this is not physically achievable due to dimensional constraints. Therefore, the practical loading capacity is **162 crates**.